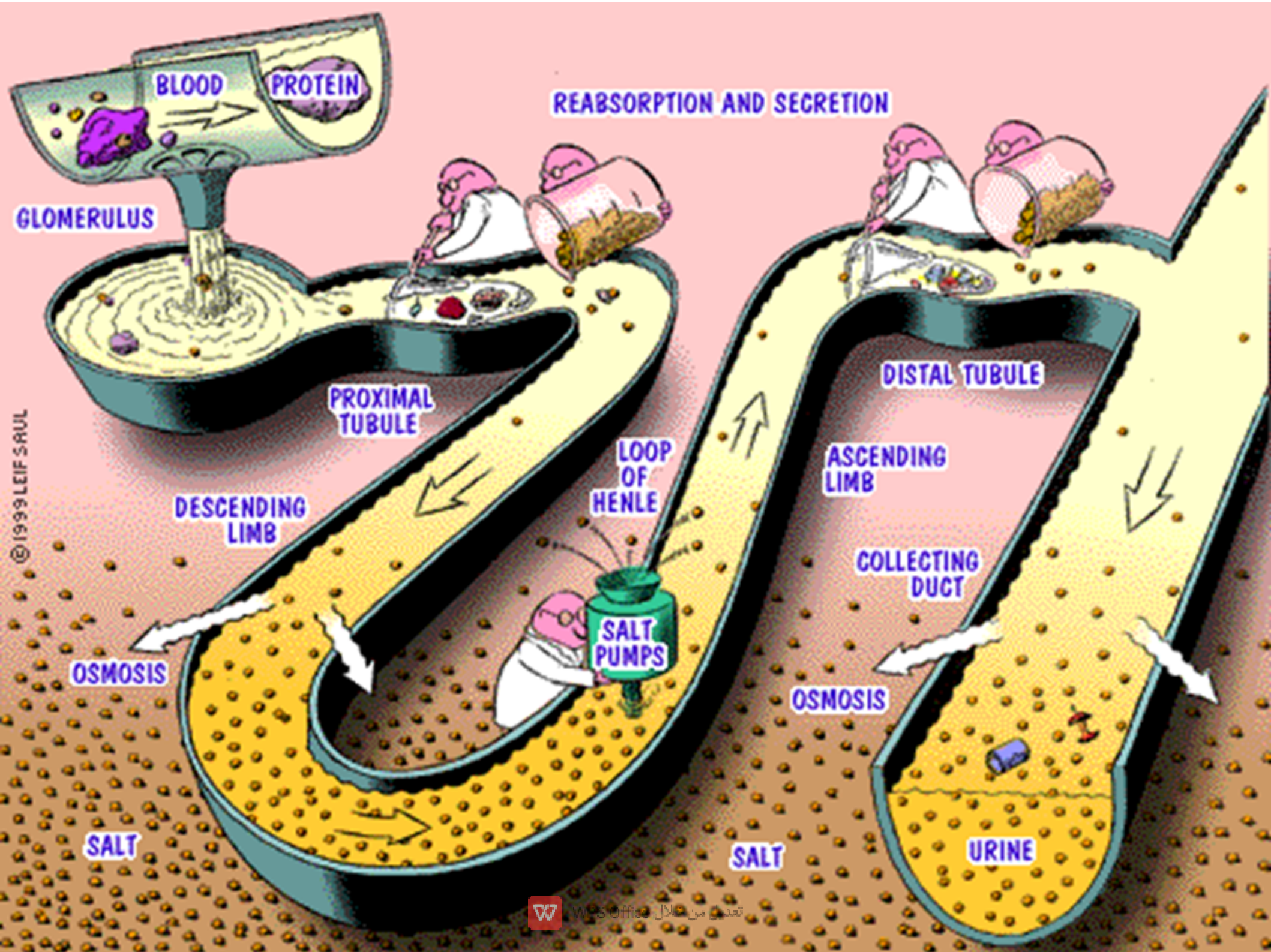




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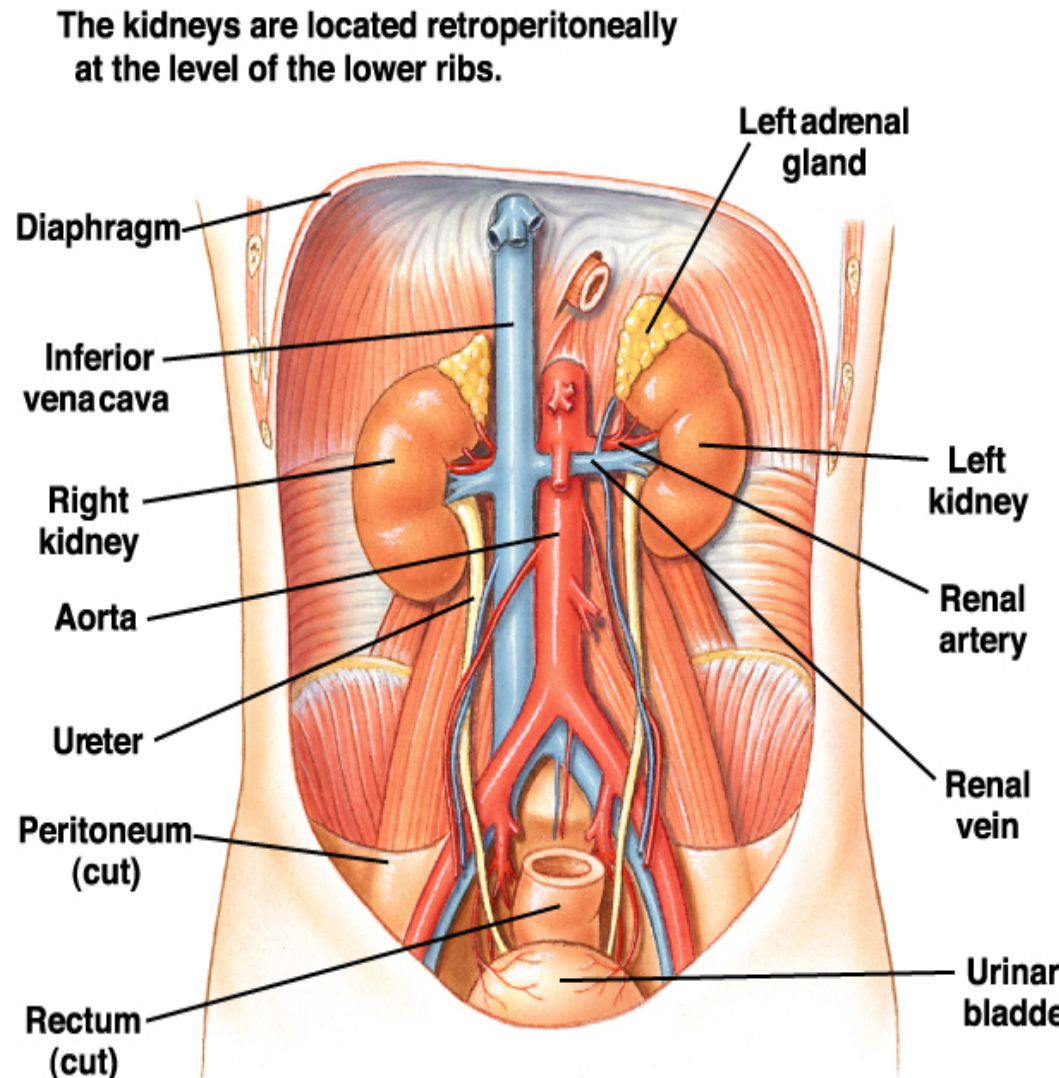
Introduction to nephrons and renal hemodynamics

Dr. Wafa Mubarak



Urinary System structures

1. **2 kidneys:** from urine.
2. **2 ureters:** transports urine from the kidneys to the bladder
3. **Urinary bladder:** provides a temporary storage reservoir for urine
4. **Urethra:** transports urine from the bladder out of the body

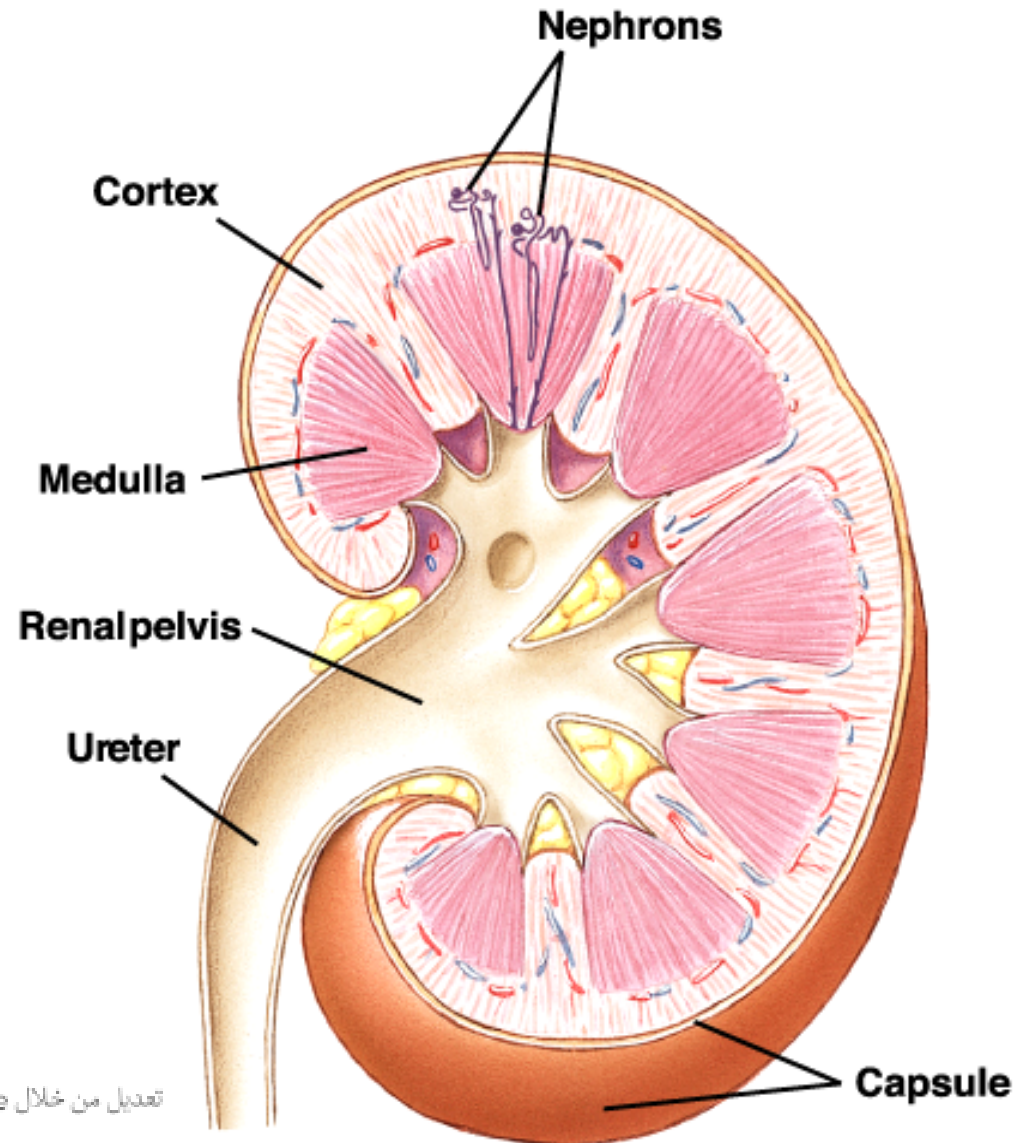


Urinary System structures: the kidney

- A frontal section shows 2 regions:

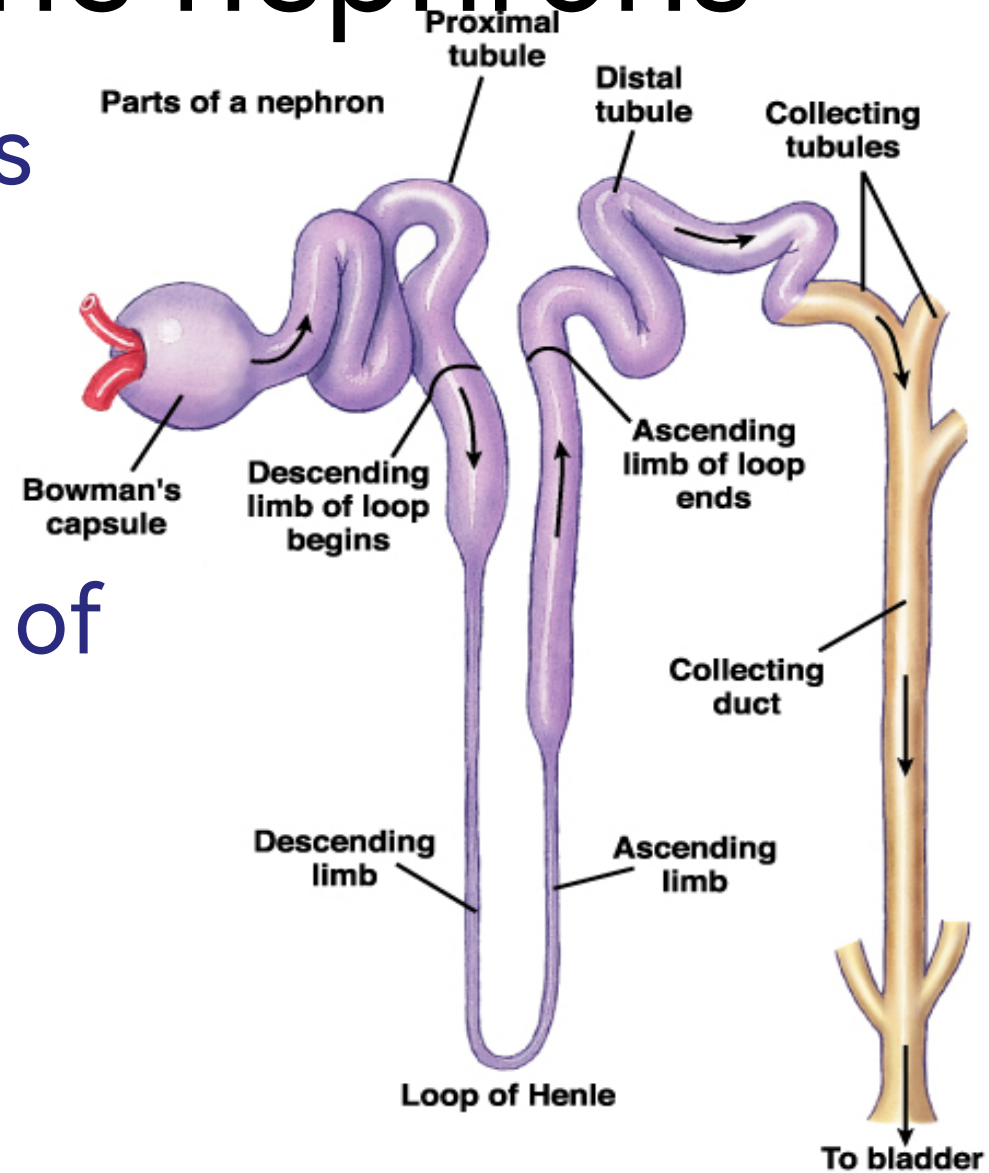
1. **Cortex:** the light colored, granular superficial region

2. **Medulla:** exhibits cone-shaped medullary (renal) pyramids



Structure of the nephrons

- Nephron is defined as the structural and functional unit of kidney.
- Each kidney consists of 1 to 1.3 millions of nephrons.
- Nephron consist of:
 1. Glomerulus.
 2. Renal tubules

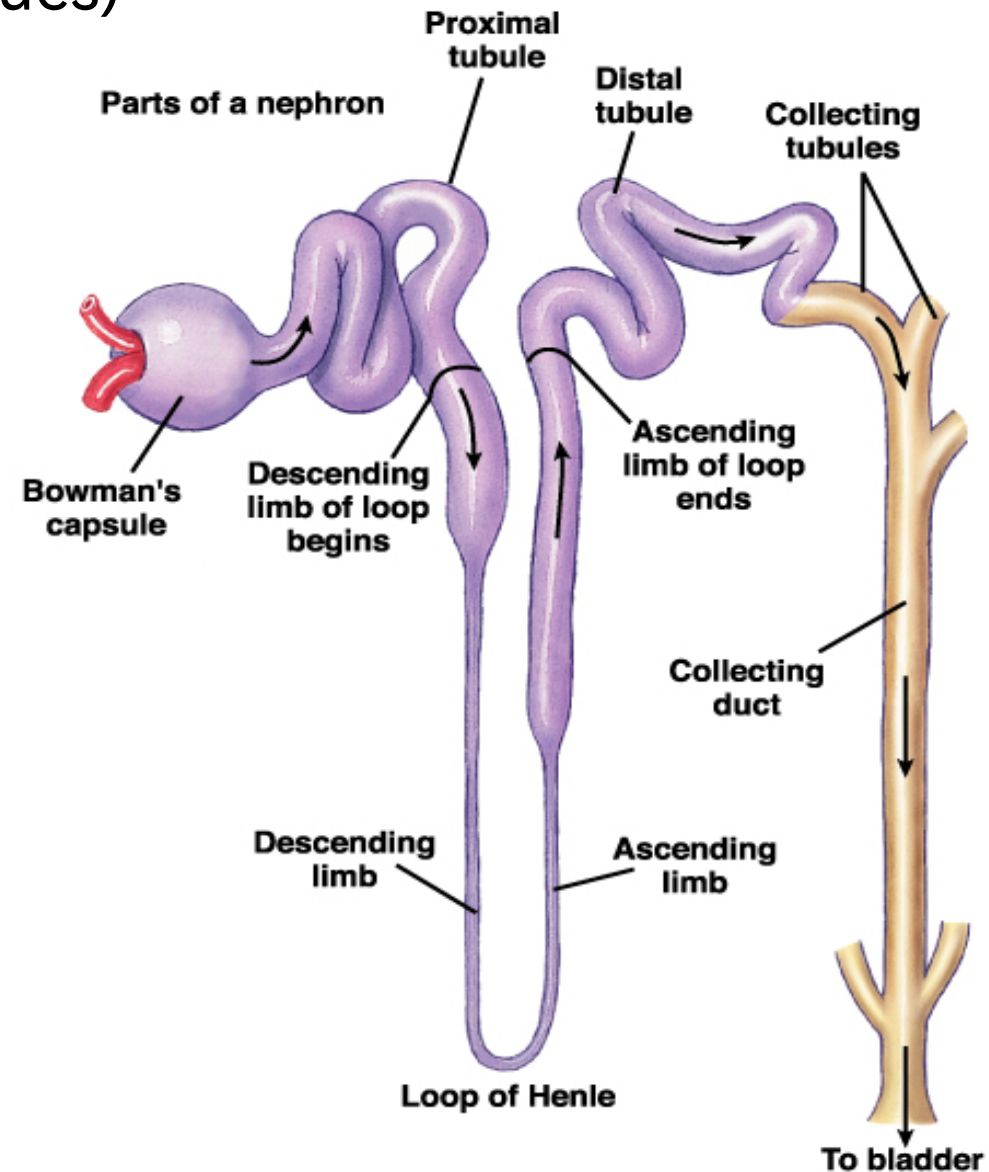


Structure of the nephrons

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- **Glomerulus:**

- a tuft of capillaries
- **Bowman's capsule:** blind, cup-shaped end of a renal tubule that completely surrounds the capillaries

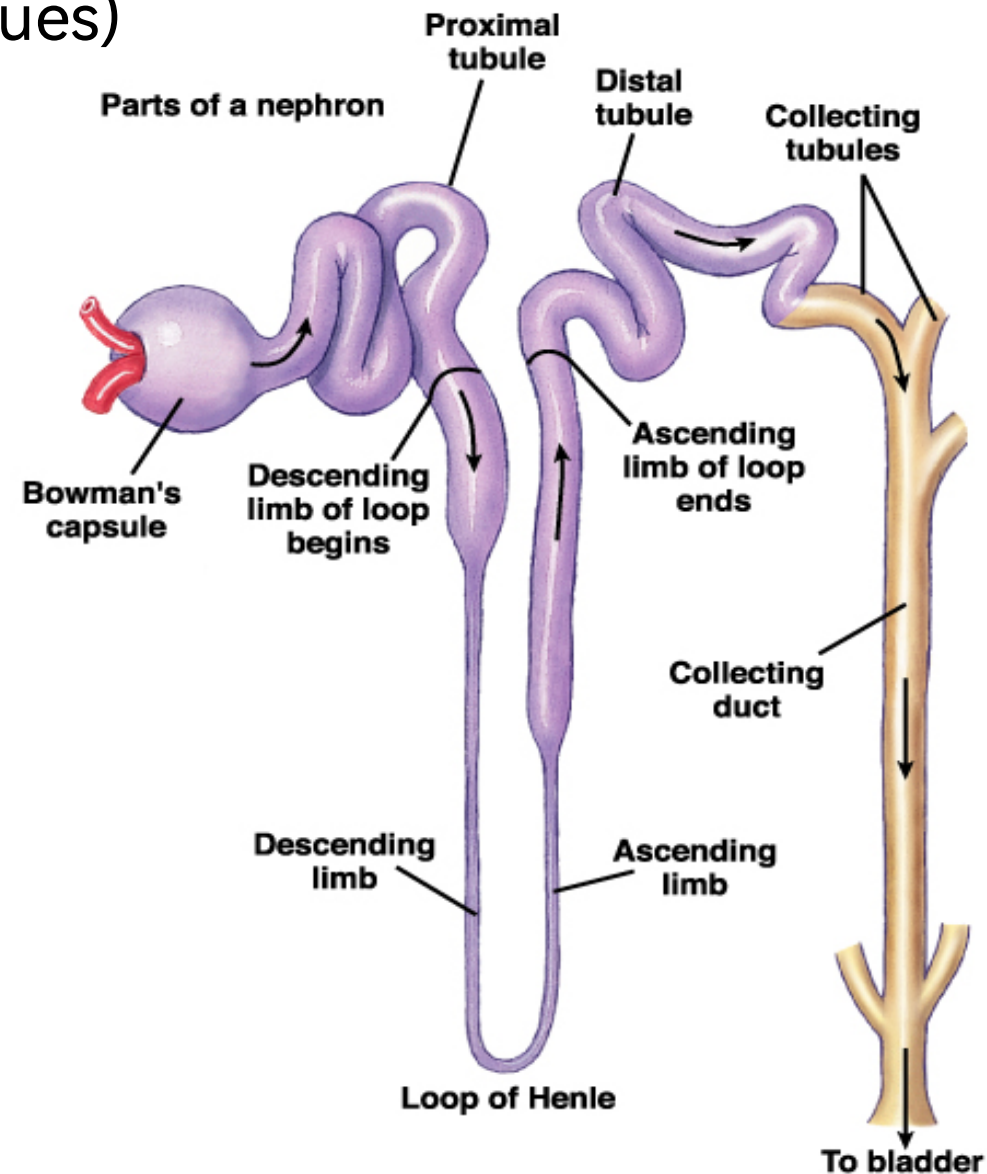


Structure of the nephrons

(continues)

2. Renal tubules:

- Proximal **C**onvolutated Tubule (**PCT**) .
- Loop of **H**enle (**LH**).
- Distal **C**onvolutated Tubule (**DCT**).
- Colleting **D**ucts (**CD**).



FUNCTIONS OF KIDNEY

- Excretion of Waste Products:
 - a. Urea (end product of amino acid metabolism)
 - b. Uric acid (end product of nucleic acid metabolism)
 - c. Creatinine (end product of metabolism in muscles)
 - d. Bilirubin (end product of hemoglobin degradation)
 - e. Other substances such as toxins, drugs, heavy metals

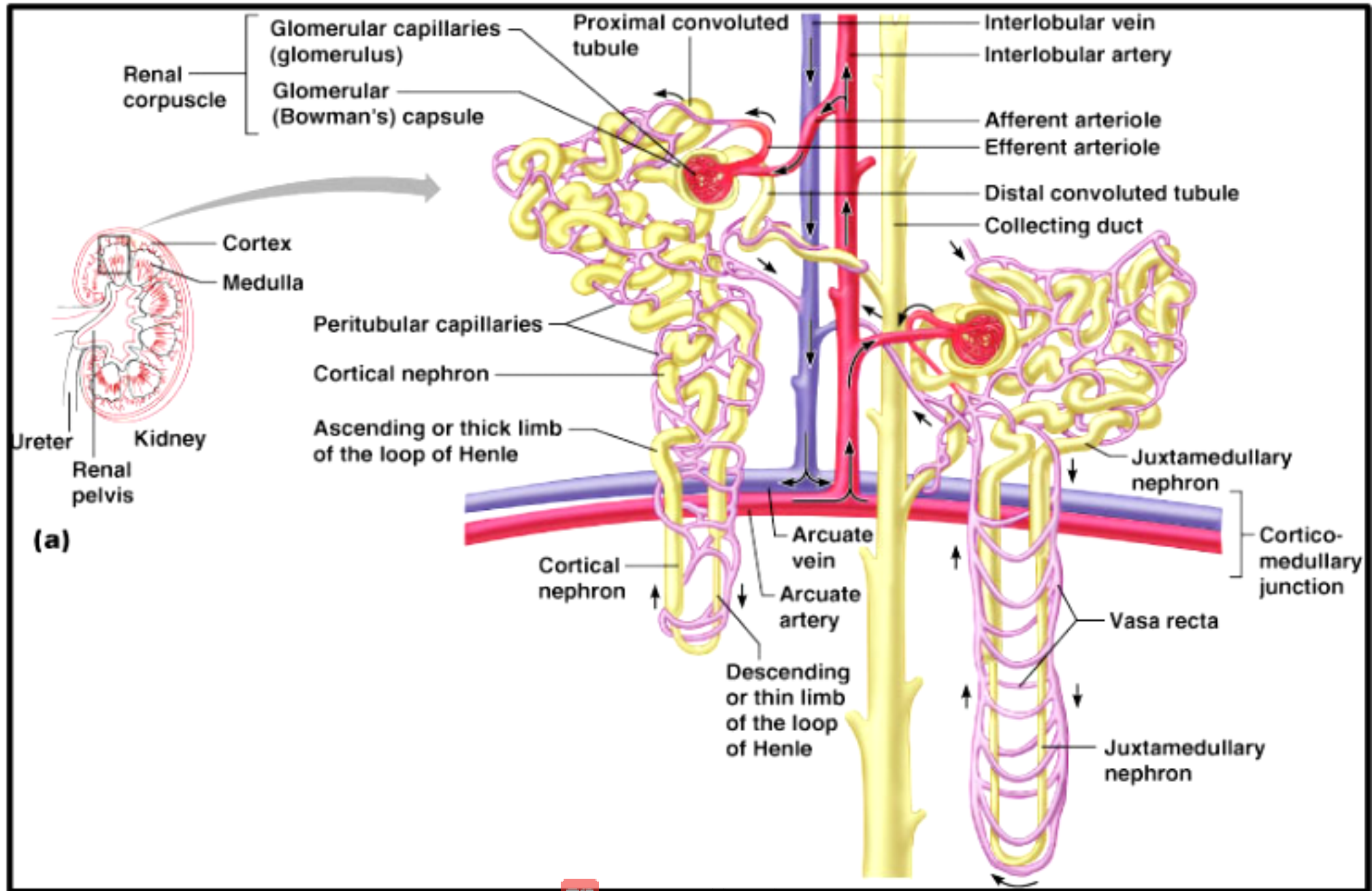


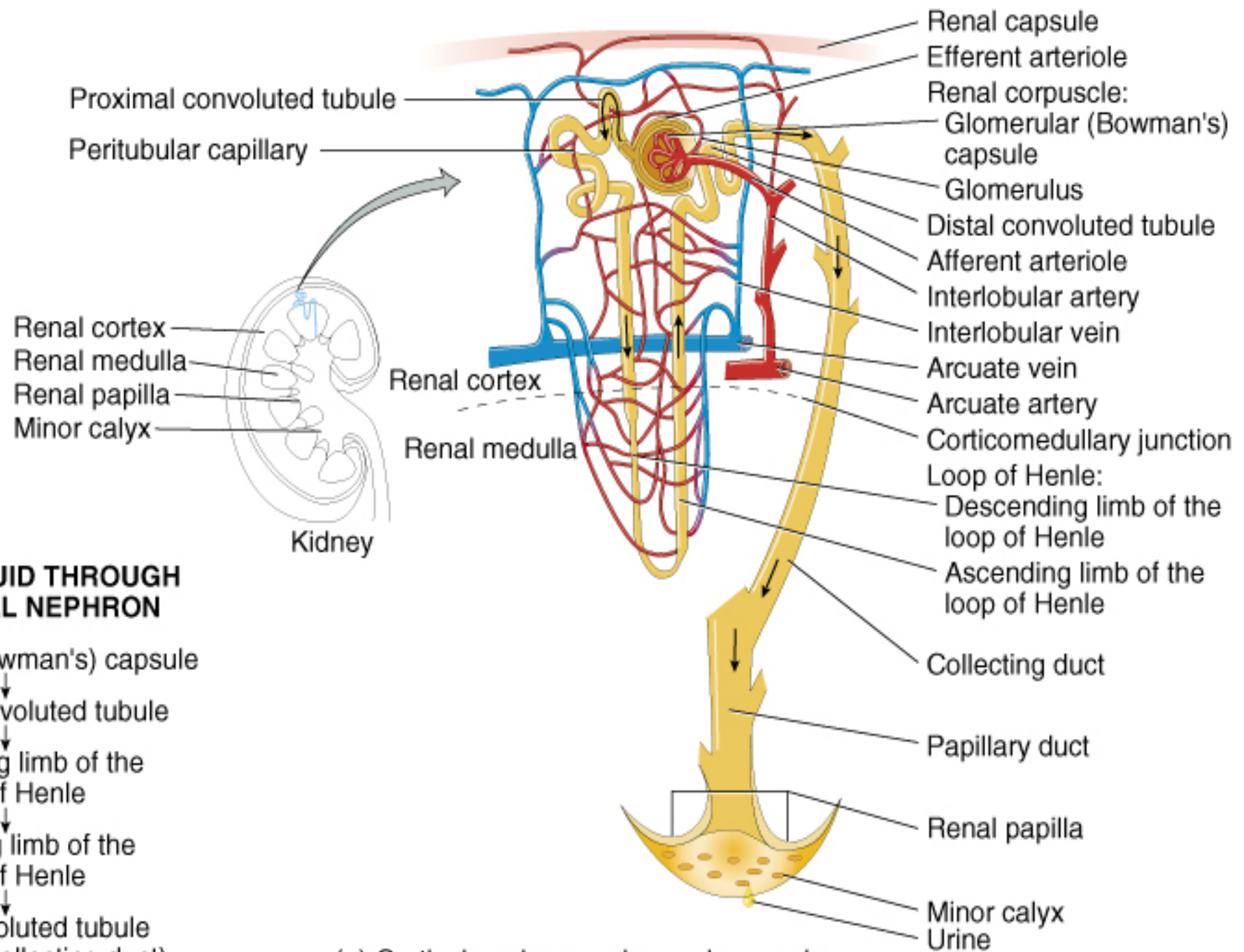
FUNCTIONS OF KIDNEY .. Continue

- Maintenance of Water Balance
- Maintenance of Electrolyte Balance
- Maintenance of Acid–Base Balance
- Regulation of blood pressure
- Regulation of blood calcium level
- Hemopoietic and endocrine function

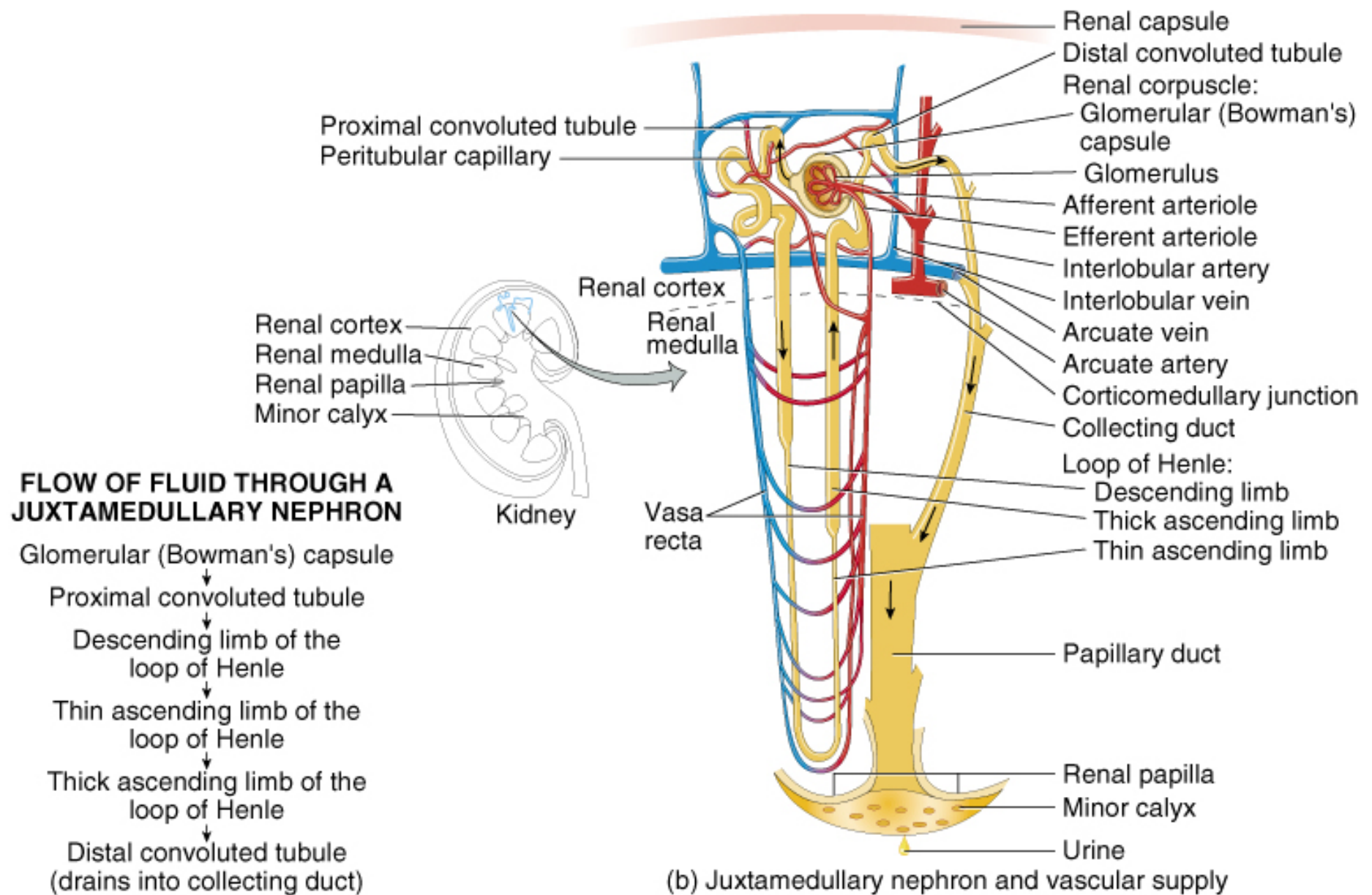


Types of nephrons





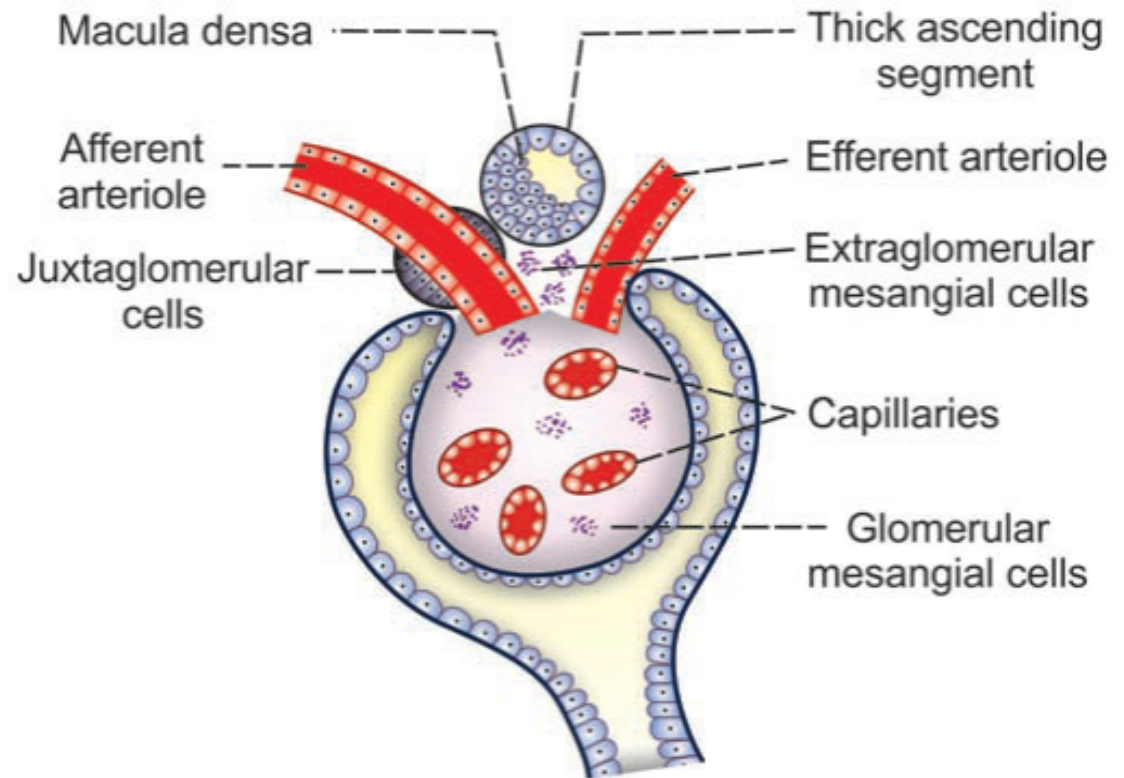
(a) Cortical nephron and vascular supply



Juxtaglomerular Apparatus (JGA)

Is formed by :

1. Macula densa
2. Extraglomerular mesangial cells
3. Juxtaglomerular cells.



Juxtaglomerular Apparatus (JGA)

- Formed where the distal tubule lies against the afferent arteriole and composed of 3 types of cells
 1. Arteriole walls have juxtaglomerular (JG) cells
 2. Distal tubule form Macula densa
 3. Mesangial cells appear to control the glomerular filtration rate



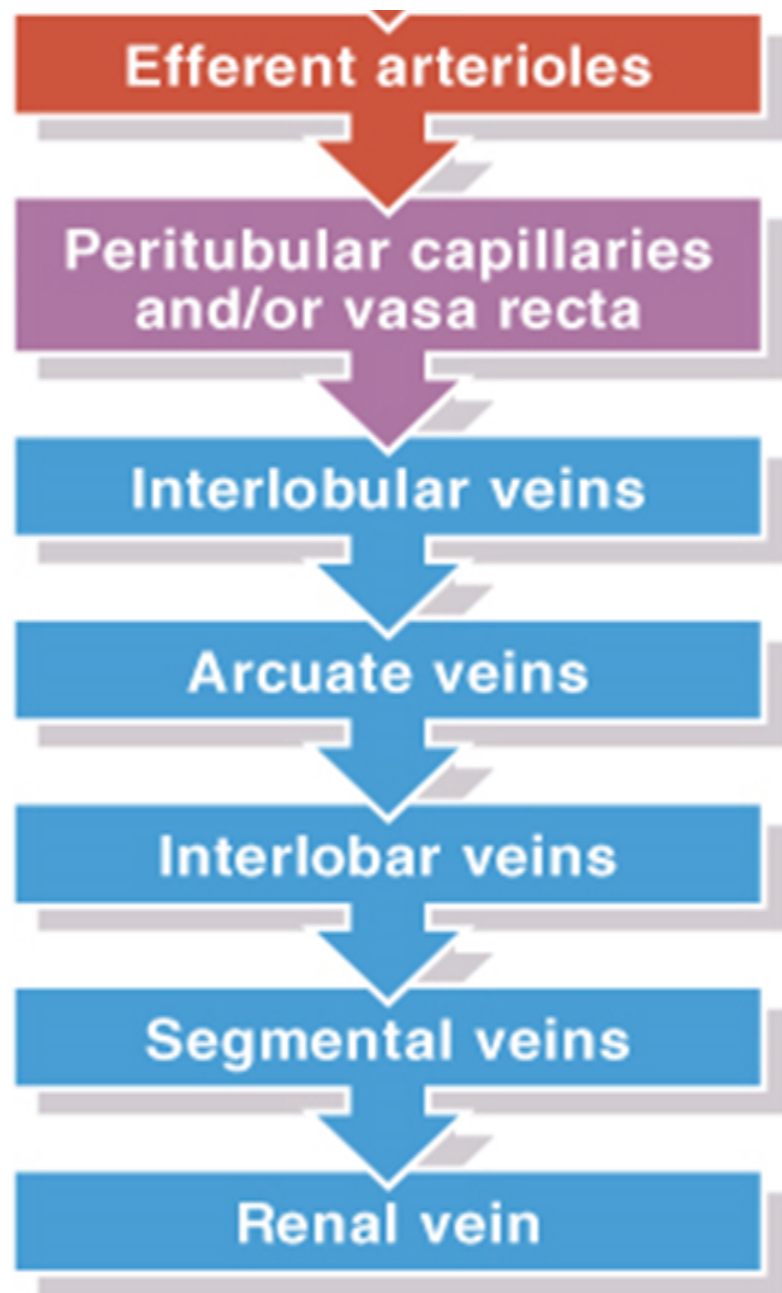
FUNCTIONS OF JUXTAGLOMERULAR APPARATUS

- Secretion of hormones.
 1. Renin
 2. Prostaglandin.
- Regulation of the glomerular blood flow and glomerular filtration rate.

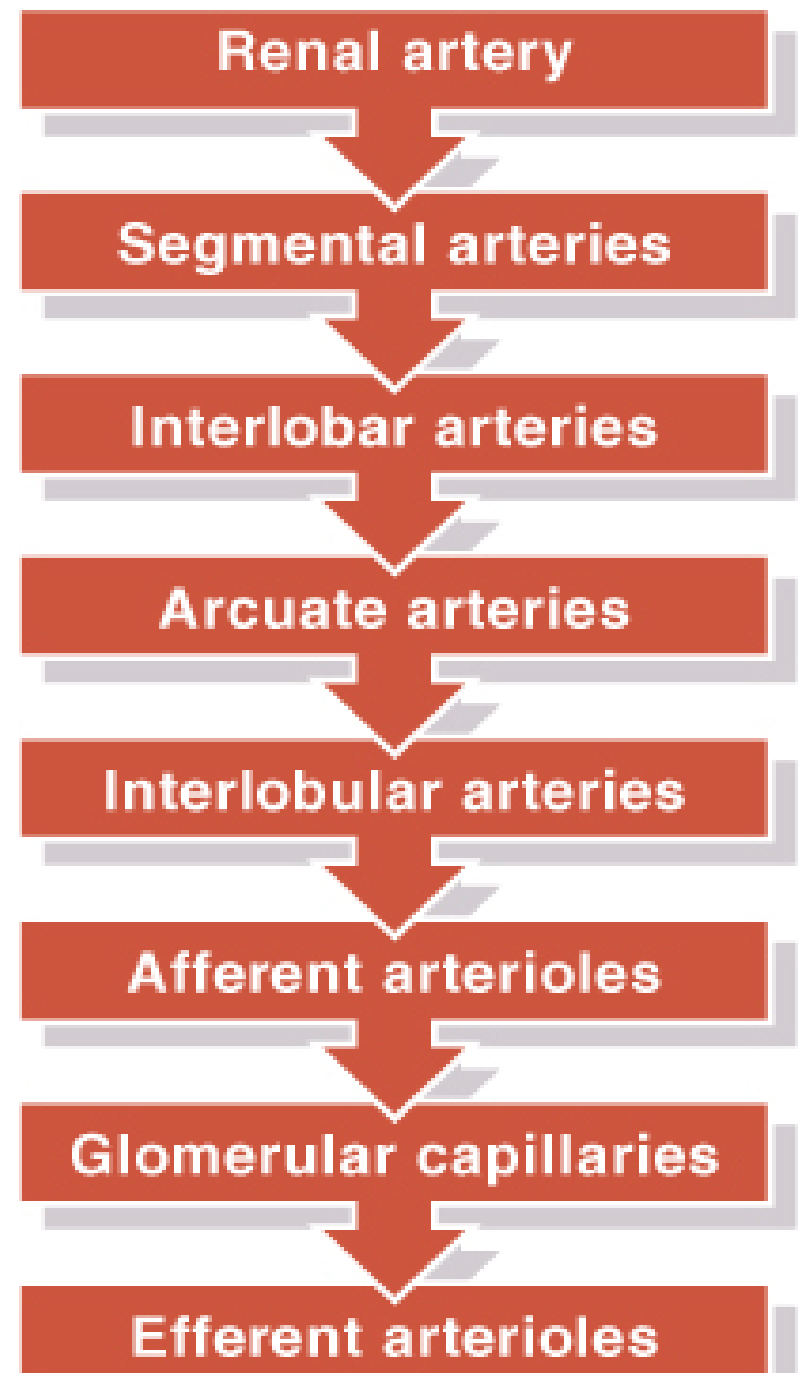


Blood Supply

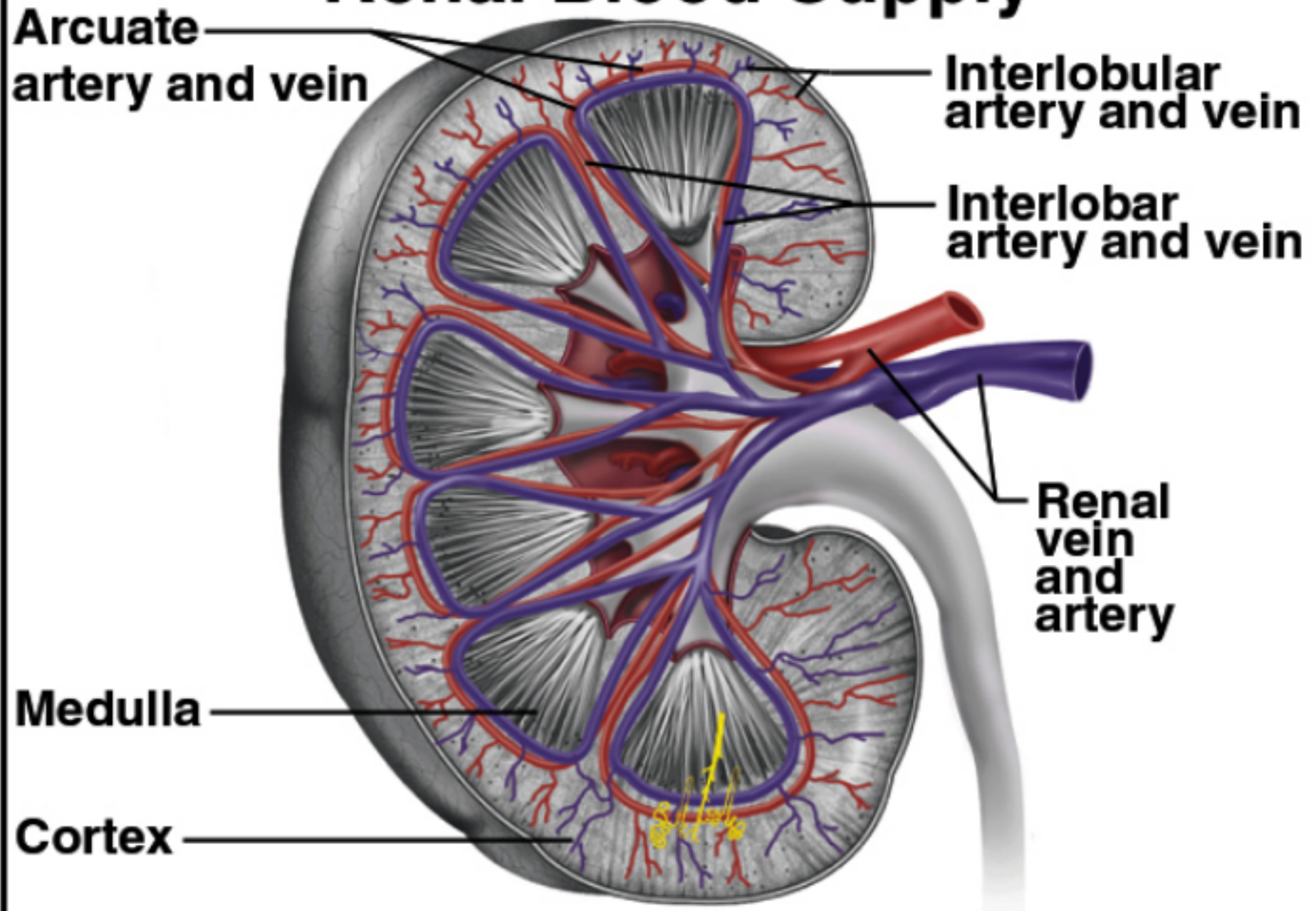
- Blood supply essential for function.
 - **Standard pathway**: blood comes in circles medulla and leaves the kidney.
 - Renal artery -> interlobar artery -> arcuate artery -> interlobular artery -> afferent arteriole -> glomerulus -> efferent arteriole -> peritubular capillaries -> interlobular vein -> arcuate vein -> interlobar -> renal vein.
 - **Vasa recta** -- efferent arteriole descends into medulla to create capillary network (peritubular capillaries) that empties into arcuate or interlobular veins.

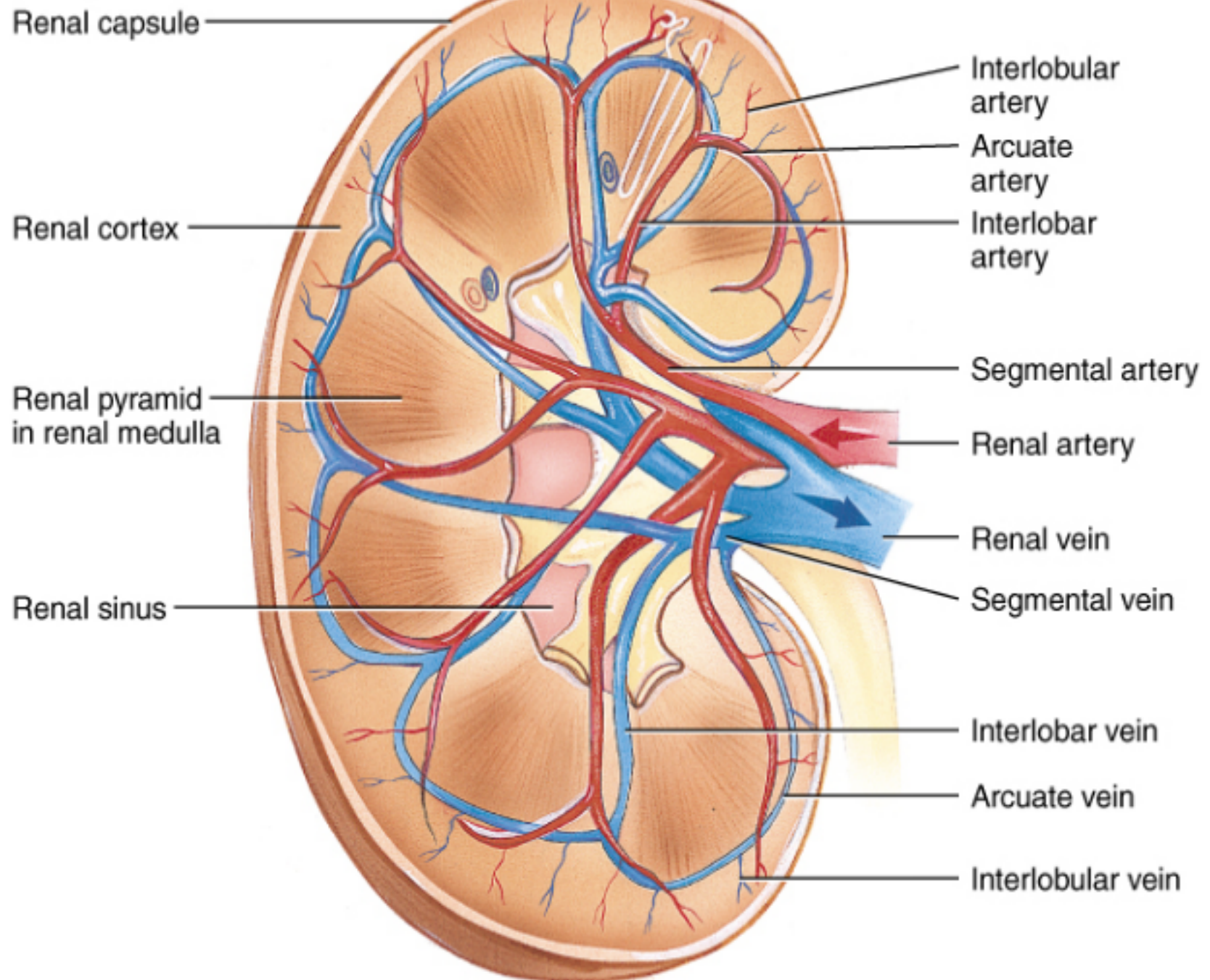


(b) Path of blood flow



Renal Blood Supply





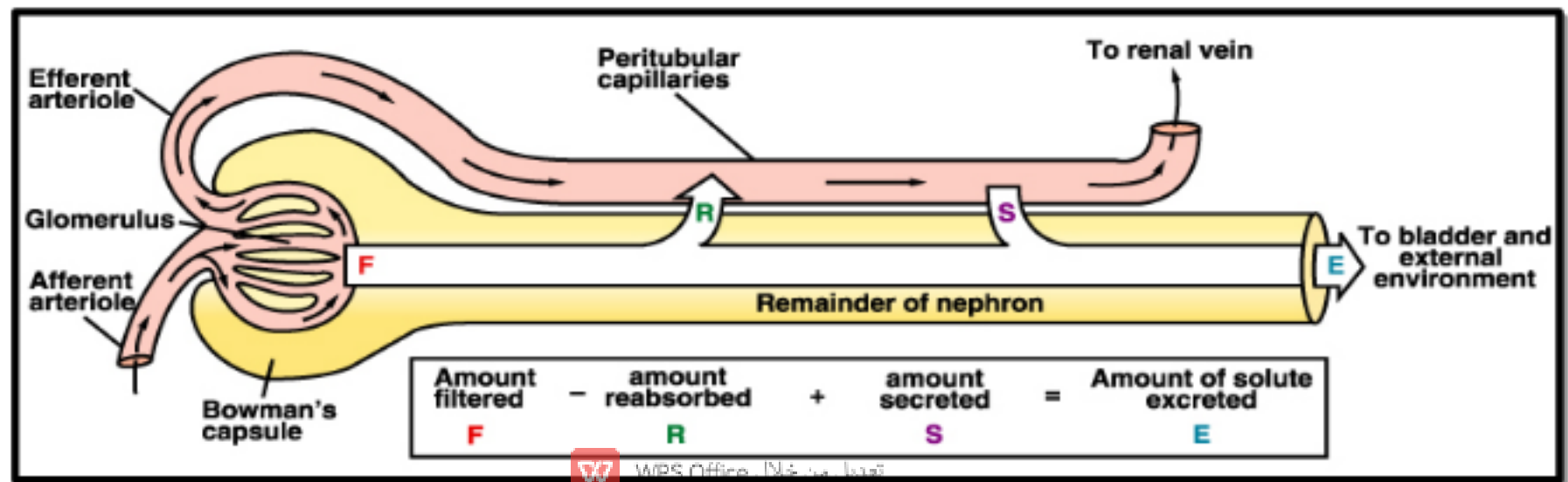
(a) Frontal section of right kidney

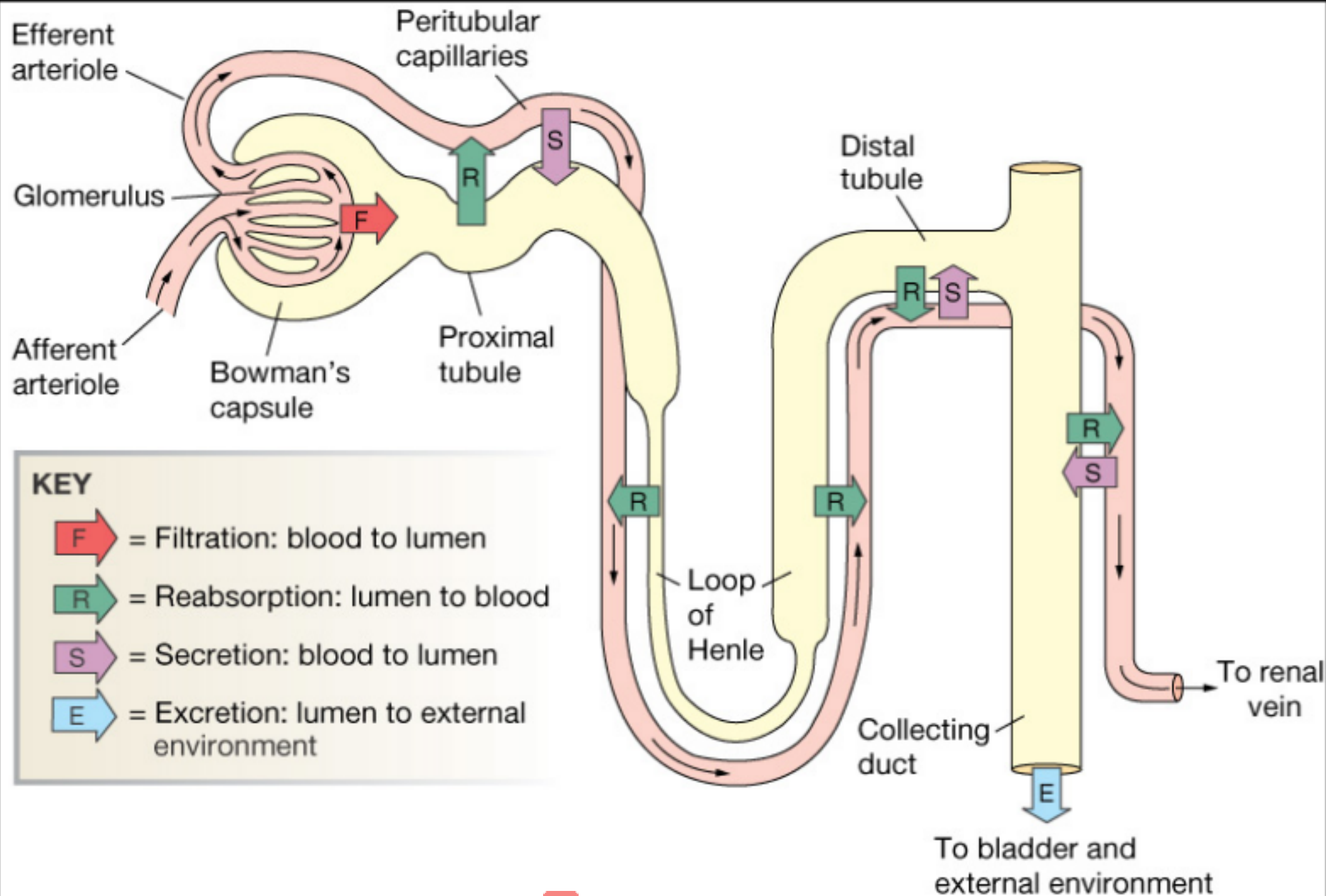
- during resting conditions both the kidneys receive 1,300 mL of blood per minute or about 26% of the cardiac output.
- Renal blood flow is regulated mainly by autoregulation .



Mechanism of Urine Formation

- Urine formation involves **three** major processes
 1. Glomerular **Filtration**
 2. Tubular **Reabsorption**
 3. Tubular **Secretion**
- the remaining part is **Excreted** as urine.





Any QUESTION

THANK YOU

